H&E Whole-Slide Images and Arm-Level Copy Number for PAM50 Subtype Classification: Internal Evaluation on TCGA-BRCA and External Validation on CPTAC-BRCA

1 Problem, Cohorts, and Why Copy Number

We predict the four-class PAM50 molecular subtype — Luminal A, Luminal B, Basal-like, HER2-enriched — from H&E whole-slide images (WSI), from arm-level copy-number variation (CNV), and from their fusion. Models are trained on TCGA-BRCA and validated externally on CPTAC-BRCA, a cohort never used for fitting, tuning, calibration or model selection. The prediction unit is the case: slide-level predictions are mean-pooled to the case before any metric is computed. Normal-like is dropped, and because the CPTAC subset contains no Normal-like case the two cohorts agree on the label space once it is removed.

Why copy number and not RNA. PAM50 labels are computed from the gene-expression matrix itself, so an RNA branch predicting PAM50 leaks the target by construction. A sibling experiment in this project reached 0.974 macro AUROC with a gated WSI+RNA model that is inflated for exactly that reason. Copy number is a different assay and carries no such circularity.

Why arm-level and not focal. Shallow whole-genome sequencing resolves copy number at the arm and large-segment scale, so a 39-feature arm-level assay is reachable by a cheap, clinically deployable protocol, whereas a model over the 19,755 focal GISTIC calls would not transfer to that setting. This is a design rationale, not a validated property: no shallow-WGS depth simulation, FFPE-degradation ablation or cost model exists in this work.

Three TCGA case counts appear below and each is correct in its own place. Table 1 gives the condition that defines each one.

Table 1: Cohort definitions. Each TCGA row is a strict subset of the one above it.

Case set	LumA	LumB	Basal	Her2	Normal	Total
TCGA labelled cases with arm-level CNV	499	197	171	78	36	981
TCGA non-Normal — the CNV branch's fitting set	499	197	171	78	–	**945**
TCGA with CNV, WSI features *and* a fold assignment	475	195	165	75	–	**910**
TCGA with CLAM out-of-fold predictions — every internal contrast	318	124	106	51	–	**599**
CPTAC-BRCA external cohort (378 slides)	56	17	27	14	–	**114**

The 945-case set is what the CNV branch is fit on when applied externally. The 910-case set is the multimodal training table: every case in it has CNV, WSI features and a fold position. The

599-case set carries every internal head-to-head, because it is the only set on which a held-out WSI prediction and a CNV prediction exist for the same case. The 311 cases in the gap are used for fitting in all ten folds but never land in a test column, so they have no out-of-fold prediction to score.

2 Approach

Two branches are trained separately and combined afterwards; a third family trains both modalities jointly.

WSI branch. Slides are tiled at 256×256 px at $20\times$ with no overlap, each tile encoded by the UNI2-h pathology foundation model into a 1536-dimensional embedding, and the bag aggregated by CLAM-MB with `-model_size big`, dropout 0.5 and an inverse-class-frequency weighted sampler. This is the primary modality: the fusion models start from its trained weights.

CNV branch. 39 chromosome arms, each feature the *median* gene-level $\log_2$ copy-number ratio over that arm. The five acrocentric p-arms carry no protein-coding content in the reference used and are excluded by construction, which is why 39 rather than 44 arms are modelled. The classifier is one `StandardScaler` $\rightarrow$ `LogisticRegression(max_iter=4000, C=0.1, class_weight='balanced')` pipeline, refit in process rather than stored as a checkpoint. The balanced weighting matters: HER2-enriched is 8.25% of the training set, and Section 5 shows that whether the WSI branch's decision rule is matched to this one changes its balanced accuracy materially.

Fusion, untrained. The *equal-weight probability mean* averages the two four-class probability vectors element-wise. It has no parameters and was fixed on TCGA before CPTAC was scored. Because it requires no training, it — not the WSI-only model — is the baseline every trained operator must clear.

Fusion, trained. Five operators pair the CLAM-MB branch with a two-layer tabular encoder over the 39 arms, sized at `-tabular_hidden_dim 64`, `-tabular_top_n_features 0`, `-fusion_hidden_dim 32`. All warm-start the WSI branch from the corresponding fold's WSI-only checkpoint and do not freeze it. `concat` concatenates the two embeddings; `gated` learns a sigmoid gate over a shared projection; `cross_attention` attends from slide to tabular embedding; `film_attention` conditions the *attention network's input* on the tabular vector through a rank-16 FiLM bottleneck with 0.2 modality dropout, so copy number re-ranks which patches are attended to; `coattn` runs co-attention between patch tokens and 22 chromosome tokens. A sixth operator, `residual`, is implemented but could not be trained: it needs a matched tabular-only checkpoint that no supported trainer in this repository produces. **The ladder is five operators, not six.**

3 Training and Evaluation Setup

Every internal model uses the distributed split set `splits/tcga_brca_subtyping_100` at $k = 10$, seed 1. All patches of a slide stay in one fold and no case is split across folds.

The splits are independently drawn, not partitioned. Each fold is its own 726/92/92 draw from the same 910 cases, so ten folds produce 920 test slots that collide onto 599 distinct cases: 357 tested once, 175 twice, 56 three times, 10 four times, one five times. A case tested by several folds receives the average of several models, which makes "WSI alone" a small ensemble flattered by roughly +0.01 macro AUROC. An audit confirmed no train/test leakage, and re-running under a random stratified partition returns the same verdict, but the absolute WSI figure should be read with this in mind. The same applies to all five operators, which share the split set, so the internal ladder comparison remains internally consistent.

Internal metrics are **pooled out-of-fold**: held-out predictions are concatenated across all ten folds and scored once. This is not the mean of ten per-fold metrics, which CLAM's own `summary.csv` reports; both appear below and are labelled. Macro AUROC is one-vs-rest. Confidence intervals and contrasts are paired bootstraps over the same case set — 4,000 resamples at seed 7 externally, 2,000 at seed 13 for the ladder — rejecting resamples that lose a class so AUROC stays defined.

Which internal CNV figure this report quotes. Three exist under three protocols: 0.8721 (CNV refit per CLAM fold, 599 cases), 0.8624 (`StratifiedKFold(10, seed 0)`, 599 cases), and 0.8657 ± 0.0034 (5-fold × 10 reseeds on the full 945-case set). The 0.010 spread is larger than several contrasts called significant here, so the choice is stated rather than left open. **This report quotes 0.8721 in every internal contrast**, because it is the only protocol under which both branches are out-of-fold on the same partition: for each CLAM fold the CNV model is fit on exactly that fold's training cases and applied to its held-out cases. The 0.8657 ± 0.0034 figure remains the right one for the CNV branch *as a model in its own right* — it is the only one computed on the full fitting set and the only one with a variance estimate — and enters no contrast here.

The WSI baseline and the five operators are *not* matched on optimiser settings (Table 2), so the ladder is a near-baseline rather than a matched comparison. This is reproduced as run rather than tidied away.

Table 2: Optimiser configuration, WSI-only baseline against the five operators. Read from each run's own `experiment_*.txt`.

Setting	WSI-only baseline	The five ladder arms
Learning rate	1.0076e–4 (swept)	1e–4 (rounded)
Weight decay	2.4457e–6 (swept)	2.5e–6 (rounded)
Bag weight	0.5534 (swept)	0.7 (CLAM default)
Instance loss / clustering	`svm` / on	none / off
Model, size, embed dim	CLAM-MB, big, 1536 — identical	
Folds, seed, split set, sampler	$k = 10$, seed 1, same splits, weighted — identical	

The instance loss and clustering rows are inert on the multimodal code path, which routes around that branch entirely; the learning rate, weight decay and bag weight genuinely differ.

4 Internal Results on TCGA-BRCA

Table 3: Pooled out-of-fold results on the 599 TCGA cases shared by all six trained runs. Δ is a paired bootstrap against the equal-weight mean, 2,000 resamples at seed 13.

Arm	Macro AUROC ↑	95% CI	Bal. Acc. ↑	ΔAUROC vs mean	ΔBal. Acc. vs mean
WSI only	0.8872	[0.867, 0.909]	0.6772	−0.0386 [−0.054, −0.023] sig	−0.0740 [−0.122, −0.027] sig
CNV only (39 arms)	0.8721	[0.848, 0.895]	0.6784	−0.0541 [−0.073, −0.037] sig	−0.0733 [−0.116, −0.033] sig
Probability mean (untrained)	**0.9259**	[0.910, 0.941]	**0.7513**	—	—
`concat`	0.8827	[0.859, 0.906]	0.6741	−0.0430 [−0.063, −0.026] sig	−0.0774 [−0.127, −0.027] sig
`gated`	0.8947	[0.875, 0.915]	0.6832	−0.0311 [−0.048, −0.016] sig	−0.0674 [−0.120, −0.016] sig
`cross_attention`	0.8917	[0.869, 0.914]	0.6848	−0.0342 [−0.054, −0.016] sig	−0.0665 [−0.117, −0.016] sig
`film_attention`	0.8818	[0.861, 0.903]	0.6652	−0.0438 [−0.062, −0.027] sig	−0.0858 [−0.132, −0.039] sig
`coattn`	0.8992	[0.879, 0.919]	0.6842	−0.0267 [−0.043, −0.011] sig	−0.0670 [−0.118, −0.017] sig

Three findings, and the third is why this thread exists.

The two unimodal branches are close. 0.8872 for H&E against 0.8721 for 39 chromosome arms, with overlapping intervals. A 39-feature logistic regression over arm medians is approximately as informative about PAM50 as a UNI2-h + CLAM-MB pipeline over whole-slide images.

Averaging them without training anything reaches 0.9259 macro AUROC and 0.7513 balanced accuracy — +0.039 AUROC over the stronger branch on that metric (histology) and +0.073 balanced accuracy over the stronger branch on that one (copy number). Both are significant.

Every trained operator finishes significantly below the untrained mean on both metrics. The best, `coattn` at 0.8992, sits 0.027 below it and its interval does not reach the mean's point estimate. Two operators fail even to clear histology alone: `concat` at 0.8827 and `film_attention` at 0.8818, against 0.8872.

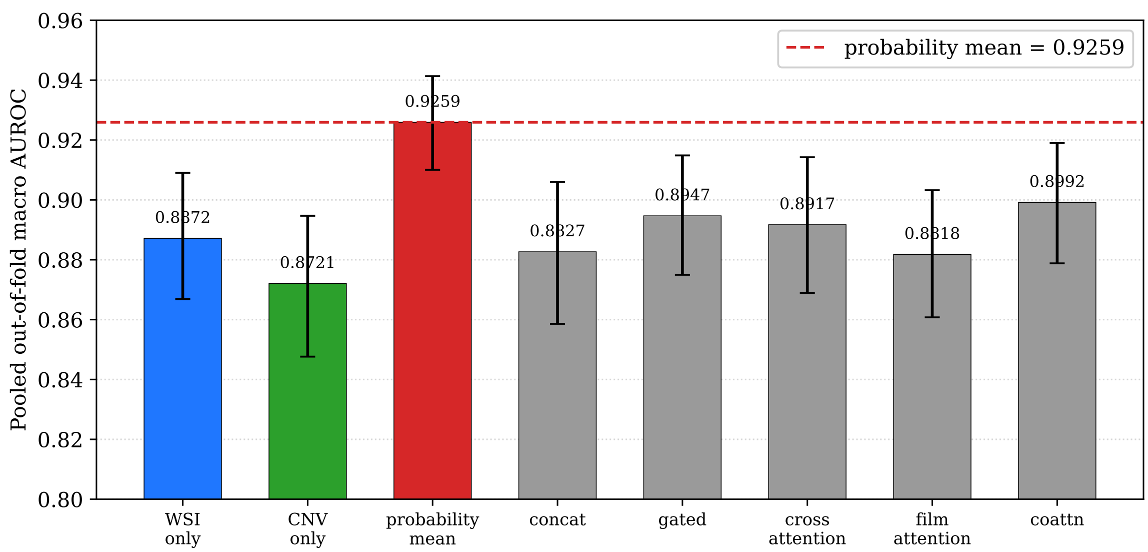

Figure 1: Pooled out-of-fold macro AUROC on the 599 shared TCGA cases, with 95% paired-bootstrap intervals. The dashed line is the untrained equal-weight mean, the baseline a trained operator has to clear. None does.

The gap is not an artifact of pooling, though it is not uniform either. Recomputing the same case-level metric within each fold's held-out cases (Figure 2), **the mean is the best arm in seven of ten folds**. In the other three a single operator edges it — `coattn` by 0.0091 in fold 6, `concat` by 0.0070 in fold 7, `gated` by 0.0039 in fold 8 — a different operator each time, on three of the four smallest test folds. That is fold-level noise, not a competitor: no operator wins twice, and none wins on the pooled set. The same figure shows the unimodal branches trading places, copy number ahead in folds 1, 2, 3 and 9 and histology in the other six.

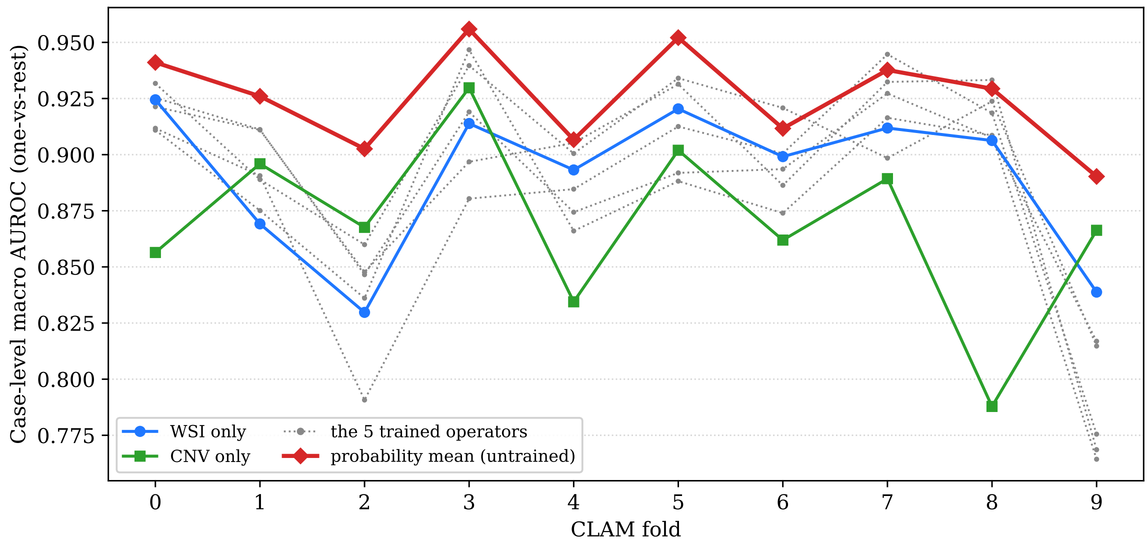

Figure 2: Case-level macro AUROC computed separately on each fold's held-out cases. The five operators are drawn undifferentiated in grey because none is consistently ahead of the others. Not the same quantity as CLAM's per-fold slide-level AUC in Table 4.

Table 4: CLAM's own per-fold held-out metrics, mean ± SD over ten folds. These are slide-level and are *not* the pooled quantity of Table 3. The CNV branch and the mean have no entry because neither is a CLAM run.

Arm	Test macro AUC ↑	Test accuracy ↑
WSI only	0.8930 ± 0.0247	0.7184 ± 0.0419
`concat`	0.8777 ± 0.0316	0.7057 ± 0.0534
`gated`	**0.8972 ± 0.0253**	0.7064 ± 0.0368
`cross_attention`	0.8936 ± 0.0324	0.7202 ± 0.0451
`film_attention`	0.8834 ± 0.0286	0.7084 ± 0.0366
`coattn`	0.8934 ± **0.0230**	**0.7330 ± 0.0354**

The two quantities do not induce the same ranking: `gated` has the highest fold-mean AUC here (0.8972) but ranks behind `coattn` on the pooled metric (0.8947 against 0.8992). These standard deviations are *across folds*, not across seeds. **Run-to-run variance has never been measured in this project** — no tolerance has been established, and bitwise reproducibility is not achievable because two operators use `nn.MultiheadAttention` with no deterministic-algorithm flags set. The only seed-to-seed figure anywhere in this report is the CNV branch's 0.8657 ± 0.0034.

5 External Validation on CPTAC-BRCA

Nothing here was refit, tuned, thresholded or selected on CPTAC. The CNV branch is fit once on all 945 non-Normal TCGA cases; the WSI branch is the frozen 10-fold TCGA ensemble applied slide by slide and mean-pooled; the fusion rule was fixed before the external set was scored.

Table 5: External CPTAC-BRCA results, 114 cases from 378 slides. 4,000 bootstrap resamples at seed 7. The fourth row is a post-hoc control, run after the raw fusion underperformed.

Model	Macro AUROC ↑	95% CI	Bal. Acc. ↑	95% CI	Her2 recall ↑
WSI only (CLAM-MB + UNI2-h)	0.8465	[0.793, 0.897]	0.5132	[0.455, 0.579]	0/14
CNV only (39 arms)	0.8883	[0.833, 0.935]	**0.7155**	[0.622, 0.805]	**12/14**
Fusion: equal-weight mean	0.9093	[0.861, 0.949]	0.6463	[0.548, 0.746]	6/14
Fusion: prior-balanced WSI + CNV *(post hoc)*	**0.9123**	[0.864, 0.952]	0.7398	[0.638, 0.832]	10/14

The operator ladder has no row here and cannot be given one. The multimodal evaluator accepts only `auto|concat|gated|residual|cross_attention`; under `auto` a `film_attention` checkpoint raises because no branch matches its parameter names, and a `coattn` checkpoint is misidentified as `cross_attention` — both build the same attention submodule — then dies in a strict `load_state_dict`. The evaluation entry point also defaults to the TCGA test split rather than CPTAC. **Section 4's ladder is therefore an internal-only result**, and the missing external column is a tooling gap, not a null finding.

Table 6: External contrasts that decide something. Every row compares two models under the same decision rule except the last, which is labelled.

Contrast	Δ Macro AUROC	Δ Bal. Acc.	Verdict
Fusion (mean) − WSI only	+**0.063** [+0.024, +0.104]	+**0.133** [+0.031, +0.240]	both significant
Fusion (mean) − CNV only	+0.021 [−0.003, +0.048]	−0.069 [−0.165, +0.022]	neither significant
CNV only − WSI only	+0.042 [−0.019, +0.103]	+0.202 [+0.089, +0.315]	AUROC ns, bal. acc. sig
Fusion (bal. *post hoc*) − CNV only	+0.0240 [+**0.00047**, +0.0499]	+0.024 [−0.055, +0.099]	AUROC sig, barely
Mixed: Fusion (bal. *post hoc*) − WSI (raw)	+0.066 [+0.026, +0.107]	+0.226 [+0.127, +0.324]	*not like-for-like*

The headline external contrast is +0.063 AUROC and +0.133 balanced accuracy, the pre-registered mean against the raw WSI branch. The last row, $+0.066/+0.226$, is arithmetically the post-hoc corrected fusion against the pre-registered WSI branch — two models under different decision rules — and appears only so a reader who meets it elsewhere can see what it is.

Fusion's edge over copy number alone is marginal, and copy number is reported on every line where a fusion number appears. Against the raw mean it is +0.021 AUROC with an interval crossing zero, and −0.069 balanced accuracy: the equal-weight mean is *worse* at decisions than copy number alone, though not significantly. Against the post-hoc variant it is +0.0240, whose lower bound is printed unrounded at +0.00047 because the customary +0.000 leaves the significance of the central contrast undeterminable. It is strictly above zero, so significant — by less than half a thousandth of an AUROC point.

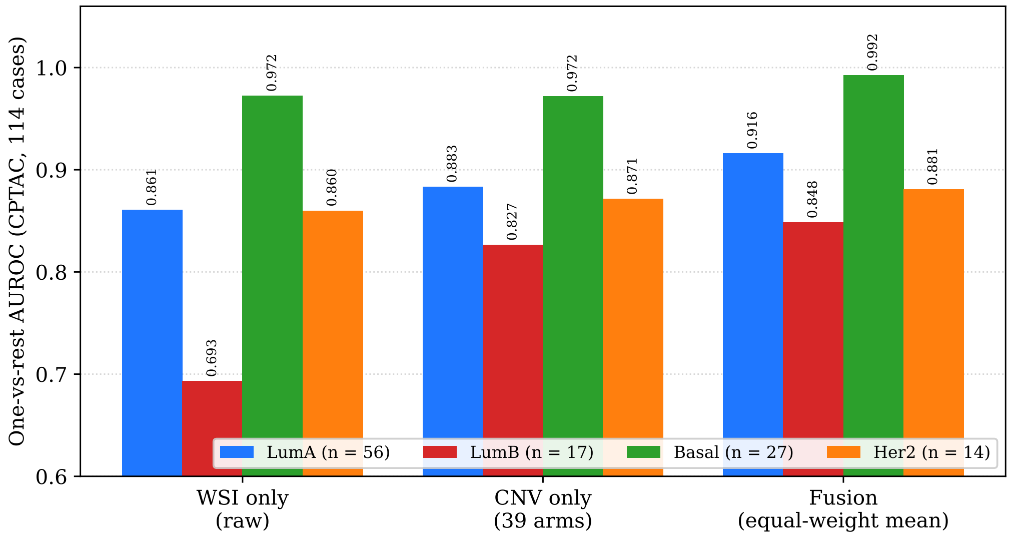

Figure 3: Per-class one-vs-rest AUROC on CPTAC. Copy number's largest advantage over histology is Luminal B (0.827 against 0.693). Per-class external estimates rest on small supports and are indicative: Her2 $n = 14$, LumB $n = 17$.

5.1 The HER2-enriched Collapse Is Real and Is Not Calibration

The WSI branch has a HER2 AUROC of 0.860 on CPTAC and a HER2 recall of 0 of 14. That combination looks exactly like a threshold artifact. It is not, and the hypothesis was tested three ways. Dividing by the TCGA training prior (HER2 = 8.25%, so a $\approx 12\times$ boost) raises the maximum HER2 probability over the cohort from 0.1335 to 0.2992 and still yields **zero** calls, because no case's boosted HER2 probability becomes its largest. Saerens–Latinne–Decaestecker expectation-maximisation, fully unsupervised, independently drives the estimated CPTAC HER2 prior to **0.000** against a true prevalence of 0.123, and also yields 0 of 14. The branch retains *partial* ranking ability — true HER2 cases average roughly twice the cohort's HER2 probability — but the head is compressed so far that no monotone prior reweighting recovers a single call.

Prior correction assumes only class prevalence has changed and the class-conditional feature distribution has not, so a recall stuck at 0/14 after a $12\times$ boost is direct evidence that **the shift is not label shift**. Internally the same model calls 26 of 51. The collapse is induced by cohort transfer, and remediation belongs on the imaging side.

Averaging a HER2-blind vector into a HER2-competent one is also what drags the raw fusion to 6 of 14 from copy number's 12 of 14 (Figure 4); matching the decision rule before fusing recovers 10 of 14 and is worth +0.093 balanced accuracy, which is the entire reason the post-hoc row exists.

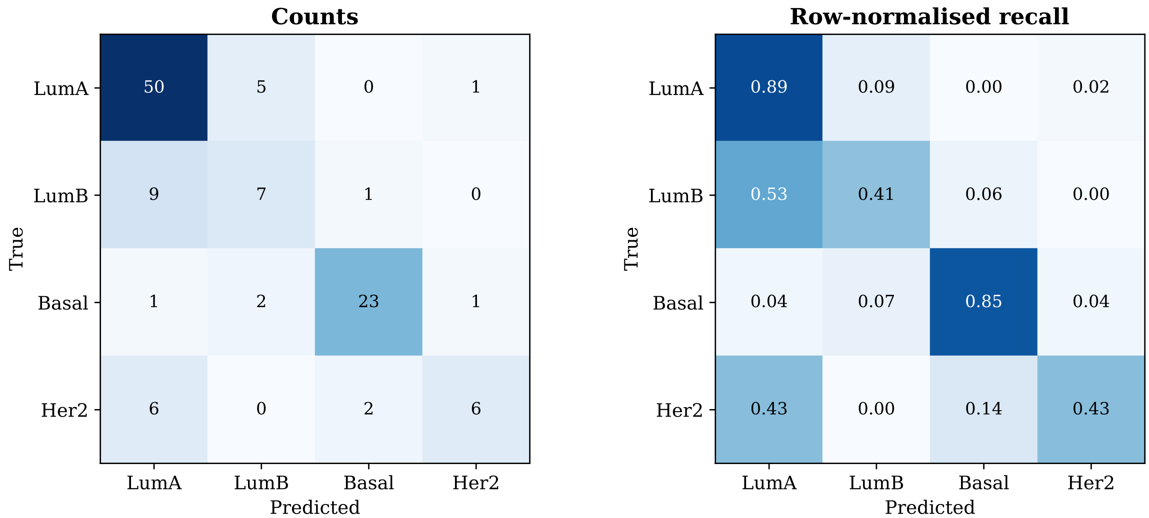

Figure 4: External confusion matrix for the equal-weight mean. Six of the fourteen HER2-enriched cases are called Luminal A, and nine of seventeen Luminal B cases are called Luminal A.

6 Why Fusion Loses to the Average

The reason a parameter-free average beats both inputs is that the branches make *different* mistakes, and the reason trained operators do not is that joint training removes that difference. Table 7 gives the error correlation ϕ between per-case correctness vectors, with the regime attached to every value because three internal ϕ figures exist and they are not interchangeable.

Table 7: Error correlation. $\phi = 0$ means the two models' mistakes are independent.

Pair	Regime	ϕ
WSI only vs CNV only	external CPTAC, 114 cases	−0.006
WSI only vs CNV only	internal, CNV refit per CLAM fold (this report's regime)	**0.193**
WSI only vs CNV only	internal, CNV via `StratifiedKFold(10, seed 0)`	0.269
The five operators, pairwise	internal, 10 pairs	**0.656** (min 0.582, max 0.706)

The mechanism claim is $\phi = 0.656$ among the five jointly trained operators against $\phi = 0.193$ between the two independently trained branches, both internal, both on the 599 shared cases. Two models trained independently on different modalities make nearly independent mistakes and averaging recovers most of the complementary signal; five operators trained jointly on a shared trunk make substantially the same mistakes, so ensembling them recovers little. Under the alternative protocol the contrast reads 0.656 against 0.269 — same direction, gap about 40% smaller. Externally the branches are effectively uncorrelated: both right on 56 of 114 cases, WSI alone on 24, CNV alone on 24, both wrong on only 10, so either-model-right is 0.912 against a WSI accuracy of 0.702.

Model count does not explain the gap. The obvious objection is that the mean averages two models while each operator is one. Ensembling all five operators reaches 0.9087, the best pair (`gated` + `coattn`, selected post hoc by maximum AUROC on the same data, a procedure that favours fusion) reaches 0.9108, and the worst pair reaches 0.8896 — against the mean's 0.9259. All three deficits are significant. **Five jointly trained models ensembled still lose to two independently trained ones.**

Learned combination rules also lose. Before building a joint head we asked whether any learned rule over the two probability vectors could beat their unweighted average. Five rules of increasing freedom — fixed mean, one scalar weight, per-class weights, and multinomial logistic regression on the probabilities and on log probabilities — were scored by nested cross-validation over the CLAM fold tags. None beat the mean on AUROC internally or externally, and every learned rule was significantly worse on external balanced accuracy. A stacker can only learn a *global* reweighting, so this says there is nothing global to learn.

`film_attention` **did not simply ignore the second modality.** Its FiLM conditioner is zero-initialised, so a collapse back to WSI-only would leave the diagnostics at zero; instead the per-case means are γ deviation 0.0161, $|\beta|$ 0.0150 and tabular logit magnitude 0.0308, all clearly off zero. It used the copy-number vector and still finished below the WSI-only branch.

One confound remains open. All five operators warm-start from the same WSI-only checkpoint, which is an obvious alternative cause of $\phi = 0.656$ that has nothing to do with joint training. Until one operator is trained from scratch, "joint training collapses diversity" and "shared initialisation collapses diversity" are not separated by anything in this repository. What can be claimed without that run is weaker but still decisive: *a trivial average of two independently trained predictors beats every trained joint fusion operator we tried, and beats all five of them ensembled together.* UniCat [8] runs the shared-initialisation control our ladder lacks and finds independent training still wins, but that is a re-identification benchmark whose numbers do not transfer and whose result reverses on one of its datasets.

7 Positioning Against Published Work

Cohort sizes, inclusion rules, splits, feature extractors, label provenance and reporting protocols differ across every row, so absolute numbers are **not** a ranking. Each comparator carries five flags: class count; prediction unit; label basis (RNA = RNA-derived PAM50, mixed, NR = not reported); regime (INT = internal CV, EXT = never-trained-on cohort, POOL = second cohort pooled into training); and R = refereed or **P** = unrefereed preprint. Where a paper reports a quantity that is not a four-class macro AUROC, the cell is - rather than converted.

Table 8: Like-for-like comparators. Numbers are verbatim from each source with its own metric.

Work	Input	Cohort	Cls	Unit/Label	Regime/Ref	Macro AUROC ↑	Note
Amer et al. [1]	gene-level CNV	TCGA n=977	4	case / RNA	INT / **P**	0.8284	the fair CNV comparator
Amer et al. [1]	CNV + WSI image	TCGA n=977	4	case / RNA	INT / **P**	0.8835	like-for-like fusion arm
Amer et al. [1]	CNV+image+clinical, *untrained* ens.	TCGA n=977	4	case / RNA	INT / **P**	0.9074	beats their own trained operators
UGES [2]	25,594 gene-level CNAs	TCGA+METABRIC	4	case / RNA	POOL / R	0.877	CNA-only ablation
UGES [2]	mutation+CNA+methylation	METABRIC→TCGA	4	case / RNA	EXT / R	0.735	genuine cross-cohort arm
PathLUPI CLAM [3]	WSI (CONCH)	TCGA→Center-2	4	case / **NR**	EXT / **P**	0.867 → 0.706	external label basis unstated
Borji et al. [4]	WSI (ResNet-18)	TCGA→CPTAC	4	**patch** / mixed	EXT / **P**	0.9523	10,000 patches/class, not cases
Fernandez-Romero [5]	WSI (UNI-2 + CLAM)	TCGA→CPTAC	**5**	slide / RNA	EXT / R	–	macro-F1 0.575 → 0.325
CustOmics [6]	WSI + RNA + CNV + methyl.	TCGA-BRCA	**NR**	case / RNA	INT / R	98.7%	RNA in the bag: target-leaking
Our work							
CNV only	39 chromosome arms	TCGA n=599	4	case / RNA	INT	0.8721	
CNV only	39 chromosome arms	→CPTAC n=114	4	case / RNA	**EXT**	**0.8883**	Her2 recall 12/14
WSI only	UNI2-h + CLAM-MB	TCGA n=599	4	case / RNA	INT	0.8872	
WSI only	UNI2-h + CLAM-MB	→CPTAC n=114	4	case / RNA	**EXT**	0.8465	Her2 recall 0/14
Probability mean	WSI + 39 arms	TCGA n=599	4	case / RNA	INT	**0.9259**	beats all 5 operators
Probability mean	WSI + 39 arms	→CPTAC n=114	4	case / RNA	**EXT**	0.9093	

Four points that a reviewer will otherwise raise. **Amer's internal 0.8284 must not be differenced against our external 0.8883** — they sit on opposite sides of a transfer boundary; the defensible statement is their 0.8284 internal against our 0.8721 internal, with our external figure having no counterpart in their work. Their copy-number representation is roughly 500 times ours in feature count, at the opposite end of the assay-cost spectrum from 39 arm medians, and their strongest single modality is a clinical branch whose contents are unspecified, so if it encodes ER/PR/HER2 their fusion comparison is circular. **Borji's external 0.9523 is not a comparable quantity**: every CPTAC row in that paper is indexed at 10,000 patches per class, so its metrics are computed on a class-balanced patch set with no case-level denominator anywhere, and its external HER2 recall of 0.767 is a recall over balanced patches rather than over 14 cases. **UGES supplies the most useful published cell for the arm-level question**: its CNA-only ablation reaches 0.877 from 25,594 gene-level calls against our 0.8657 ± 0.0034 from 39 arms, which is the strongest evidence that coarsening to arm scale costs very little. **And the positioning claim survives only in its qualified form**: no published *multimodal* PAM50 model reports an external, never-trained-on, PAM50-specific evaluation. Four unimodal H&E models do, so the unqualified version is false and should never be written.

One instance of our central negative result already exists in the literature unnamed: Amer's own untrained "Simple Ensemble" scores 0.9074 in the CNV+image+clinical combination, above both of their trained operators, and their text concedes it in passing. Their paper never defines what that baseline computes, so it should be described as an untrained ensemble and not as a probability mean.

8 Outcomes and Next Steps

8.1 What this thread established

1. **An externally validated multimodal PAM50 result exists, and it is 0.9093** [0.861, 0.949] on 114 CPTAC cases, against 0.8465 for histology alone. The gain over histology is significant on both AUROC (+0.063) and balanced accuracy (+0.133). No published multimodal PAM50 model has an external evaluation of any kind, so this has no direct comparator.

2. **The honest headline is copy number, not fusion.** A 39-feature logistic regression is statistically indistinguishable from a UNI2-h + CLAM-MB pipeline on an independent cohort (ΔAUROC +0.042, interval crossing zero) and significantly better at the decision level (Δ balanced accuracy +0.202). Fusion's edge *over copy number alone* is +0.021 AUROC, not significant, and negative on balanced accuracy. Reporting fusion without the CNV-alone arm would reproduce the selective reporting the literature survey criticises.

3. **Trained fusion loses to an untrained average, and the mechanism is measured.** All five operators finish significantly below the equal-weight mean, two below histology alone, five ensembled together still below two independent models, and the error correlation is 0.656 among jointly trained operators against 0.193 between independent branches. No published WSI+omics work benchmarks a fusion operator against an equal-weight probability mean, and none reports an error diversity statistic as the mechanism.

4. **The HER2-enriched collapse under cohort transfer is real and is not a calibration artifact.** 0/14 raw, 0/14 after a 12× prior boost, 0/14 under unsupervised SLD-EM, against

26/51 in domain. Prior correction is exactly the remedy for prevalence shift and it fails completely, which localises the problem to the imaging side.

8.2 Gaps to target

- **No external evaluation of the operator ladder**, because two of five checkpoints cannot be loaded by the evaluator and it defaults to the wrong split. Every ladder conclusion is internal.
- **The warm-start confound is open.** All five operators share an initialisation, so $\phi = 0.656$ may be shared init rather than joint training.
- **Run-to-run variance is unmeasured**, so no tolerance is quoted anywhere and bitwise reproducibility is neither claimed nor achievable.
- **Preprocessing is not held constant across cohorts.** Geometry matches (256 px at 20×, 1536-dim), but CPTAC features carry CLAM tiling attributes and TCGA features carry Trident ones, so tissue segmentation — and which tiles enter each bag — differs. This is a live confound for the domain-shift account of the HER2 collapse.
- **CPTAC tissue preservation type is unrecorded.** Whether the 378 slides are FFPE, frozen or mixed is established nowhere in this project, and published CPTAC-BRCA subsets differ on exactly that axis.
- **External power is low for two classes**: Her2 $n = 14$, LumB $n = 17$.
- **The 39 orphan cases and 112 unlabelled slide sets** on disk cannot be recruited: 35 cases have CNV but no slide (already used by the CNV-only arm), and 112 cases have slide features but neither a PAM50 label nor copy number.

8.3 Recommended next steps, in priority order

1. **Train one operator without the warm start** (≈2.5 GPU-hours, one job). This is the highest value remaining run: it converts the mechanism claim from "joint training or shared initialisation" into one or the other, and it is the one open objection to the strongest result in this thread.
2. **Extend the evaluator to `film_attention` and `coattn` and point it at CPTAC** (code change, no retraining). This gives the ladder its missing external column and tests the operator-versus-mean comparison under transfer, where the branches' errors are even less correlated ($\phi = -0.006$ against $+0.193$ internally).
3. **Multi-seed the CNV branch inside the external comparison** (seconds of compute). The external numbers currently rest on a single fit.
4. **Resolve the CPTAC preservation type** from PathDB/TCIA metadata before the domain-shift attribution is finalised. More specific and more checkable than the staining hypothesis.
5. **Stain normalisation and encoder ablations on the WSI branch**, since the HER2 collapse is shift-induced. Note that at least one independent group deliberately omits stain

normalisation on the grounds that it may impair foundation-model generalisation, so this is a candidate remediation rather than an agreed one.

6. **Consider regenerating the splits as a true 10-fold partition.** This would raise the internal comparison set from 599 to 910 cases, a 52% gain in evaluation power, but requires retraining the baseline and all five operators and would invalidate the CPTAC inference. It would also remove the $\approx +0.01$ ensemble inflation, so the WSI figure would fall.

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